Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/28/2017
PI Name:	Roux, Stanley Ph.D.		
Project Title:	SporeSat 2 Mission: Investigating Biophysical Mechanisms of Plant Gravisensing Using a Lab-On-A-Chip Approach (CoPI=Roux)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture (2) Bioregenerative Life Support		
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Zip Code:	78712-1240	**Congressional District:**	25
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	11/01/2014	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	Small Satellites		
Flight Assignment:	NOTE: Extended to 12/31/2016 (original end date 8/31/2015, subsequently extended to 12/31/2015 and then 6/30/2016), per NSSC information (Ed., 5/27/16) NOTE: Extended to 6/30/2016 (original end date 8/31/2015 and then extended to 12/31/2015), per PI and NSSC information (Ed.,1/11/16) NOTE: Extended to 12/31/2015 (original end date 8/31/2015), per PI and NSSC information (Ed., 8/18/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Rickus, Jenna Ph.D. (CoPI: Purdue University)		
Grant/Contract No.:	NNX15AB85A		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Ed. NOTE (June 2015): Continuation for SporeSat 2 of "Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach"--grant NNX11AF48A.
SporeSat is a Small Complete Mission based on a Mission of Opportunity-2 nanosatellite designed for real time measurement of calcium signaling and maintenance of variable gravity treatments during flight. The impetus for SporeSat is to gain a deeper knowledge of the molecular and biophysical mechanisms for plant gravity sensing by studying the activation of cell gravity-sensing and electrophysiological signaling in the single-cell model system, Ceratopteris richardii, using a novel ‘lab-on-a-chip’ microsensor technology platform called a CEL-C advanced bioCD.

In 2013, SporeSat was adapted into a 2 flight mission strategy with flight 1 focused on the technology objectives and flight 2 focused on the scientific objectives. In April 2014, SporeSat-1 was deployed from a SpaceX Falcon 9 launched from Cape Canaveral as part of CRS-3. SporeSat-1 was successfully commanded and science data downloaded.

In late 2012, the original SporeSat Principal Investigator (PI), Dr. Marshall Porterfield, was tapped as the director of NASA’s Space Life and Physical Sciences Research and Applications Division. Because SporeSat falls under this division at NASA, due to conflict of interest, Dr. Porterfield was replaced as PI with his close research collaborator and fellow expert in physiological sensing and biosensor technology, Dr. Jenna Rickus. Due to the increased emphasis on the Science Objectives for SporeSat-2, Drs. Stan Roux (original co-Investigator) and Rickus will serve as Co-Principal Investigators for SporeSat-2. Dr. Rickus will lead all aspects of the bioCD design, performance, and production at Purdue and Dr. Roux (at University of Texas Austin) will lead all biological aspects of the project particularly that related to experimental design and interpretation of the experimental data as well as setting the science requirements for the SporeSat-2 mission.

At the time of loading for SporeSat-1 the Hn-n Ceratopteris richardii spores were not available; therefore, Rn5 spores from the University of Texas (UT) were used. In the SporeSat 2 mission we will use the now available Hn-n spore strain that was used in all previous calcium current studies prior to SporeSat. Using the Hn-n from UT Austin and the upgraded SporeSat-2 hardware we will record ground based calcium currents in stationary positions (variable = angle off gravity) and under rotation (variable = artificial gravity magnitude). In addition, as mutant spores become available we will integrate the mutants into the experiments.

The SporeSat-2 experimental paradigm and geometry is different from any previous experimental set ups. As in any biosensor experiment, the Ca2+ ISEs are sampling at a particular location in space and time, a dynamic calcium current that exists across the 3D surface of the fern spore. To assist in development of the biological model based on sensor recordings, we have developed a 3D simulation model that predicts the 3D calcium concentration profiles in the SporeSat wells as well as the sensor recordings over time based on the biological flux from the fern spore. We will therefore be able to compare our experimental data with our predictions for different conditions (angle off gravity, under rotation) as well as for different calcium mutants.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Astronaut health in long term space missions is likely to be dependent on and benefit from plants. On Earth we depend on plants for food, clean air, and waste treatment. The same will be true in space. If we as humans are to achieve the kind of space exploration and habitation that we all imagine and dream of, we are going to need plants up there with us. Yet on the International Space Station (ISS), gravity is less than one one-thousandth of the gravity here on Earth. On the Moon, it is about 17% of Earth's gravity. In addition, there are astonishing similarities in signaling within cells across species. As in plants, calcium plays important roles in human cell development, function, and dysfunction. Understanding gravity sensing in plant cells is likely to give clues to gravity sensing in other organisms including humans. In addition, the technology that we developed for these experiments establishes a platform to ask other questions regarding human cell responses on Earth and in space. We are already looking at how we can adapt the technology to study pancreatic beta cells and diabetes.

Summary of Research Accomplished for NNX15AB85A
Title: SporeSat Mission: Investigating Biophysical Mechanisms of Plant Gravisensing Using a Lab-on-a Chip Approach

A key objective of the SporeSat project was to produce a next generation lab-on-a-chip device to be used in microgravity to determine the minimum g force needed to activate the calcium channel that regulates calcium ion transport in spores of the fern Ceratopteris richardii. In addition, the original proposal included the objective of observing the calcium current response of gene knockout mutants. Through the first SporeSat project and the Phase one activities of SporeSat-2, we have achieved many of the Technical Objectives described in the SporeSat proposal, which we will describe here under five headings. We also add a final Significance section that summarizes the importance of the discoveries made.

Light Initiation of Spore Germination: We determined that 1 hour of light is sufficient to initiate the germination of Ceratopteris spores and allow the gravity-directed polarization process to proceed through 24 hours. This assures that the battery power planned for the light treatment in the SporeSat mission will easily suffice. Spores require an additional light exposure between 28 and 40 hours after the first light treatment to complete germination, but the calcium ion transport investigated in the SporeSat mission occurs within the first 24 hours of development. These new data provide important evidence that the photoreceptor and irradiance needed to initiate the calcium current are present during the first hour of light treatment planned.

Spore and Media Loading: Early in phase one of SporeSat-2 we found that the loading method used allowed the spores to dry after hydration, which resulted in inhibition or significant delay in germination. To solve this loading problem scientists at the Ames Research Center (ARC) implemented the use of a capillary pipette and micromanipulator for spore loading and adapted an off the shelf 3D printer to automatically load agar germination media into the wells. These improvements resulted in spore loading time of 2-4 hours, and complete filling of greater than 50% of wells without bubbles. We determined that complete loading of the well with germination media without bubbles is necessary for resolution of the BioCD signal.

Germination Media Calcium Content: We determined that the agar media with the lowest background calcium

Task Progress:	concentration that supported both normal spore germination and strong electrochemical signals from spores was Noble agar, which had a background [Ca2+] of ~280 µM, and this became the agar of choice for the successful test run we completed in April (Zero Day Test 12). Success in Detecting Gravity-Directed Electrical Signals from Spores: Thanks to the reduced background Ca2+ in the agar, improvements in spore loading, and an improvement made to the signal detection software, we have developed an effective BioCd. Dr. Rickus was able to detect statistically significant electrical signal from spore cells that increases with g-force, and is different from negative controls in Zero Day test #12. This success replicated results observed in prior Zero Day tests, but with a dramatic reduction in noise and increase in signal stability. These results, which have now been published (Park et al., 2017--see Bibliography section), indicate that spore, spore loading, and instrument conditions have been found that reproducibly confirm that the bioCD can record significant electrical signals in the spore that are induced by gravity. The likelihood that these signals are due to changes in calcium transport is very high. Mutant Plant Development: The original SporeSat proposal included the objective of generating knockout mutant spores for the Ceratopteris Ca2+ ATPase (CrACA1) and Annexin (AnnCr1 and AnnCr2) genes. At this time we have generated mutant sporophyte plants that possess short hairpin constructs designed to reduce or suppress the expression of the CrACA1 and AnnCr2 genes. In addition to these mutant plants, we have also generated mutant sporophyte plants that possess short hairpin constructs designed to suppress the expression of genes that encode a Mechanosensitive Channel of Small Conductance (CrMscS) and the ubiquitous calcium regulatory protein Calmodulin (CrCaM). We have also generated sporophyte plants that are up-regulated in the expression of the AnnCr2 and CrMscS genes. We have currently verified three plants with significant increase in transcript abundance of AnnCr2 and three plants with significant decrease in transcript abundance of AnnCr2. We have also verified that one plant has significant increase in transcript abundance of CrMscS. We have collected spores from all of these plants. Significance An elegant aspect of the SporeSat design is the inclusion of variable g force control within a microgravity experiment. The implementation of this aspect of the design was demonstrated in the first SporeSat mission, and this success will help assure the success of SporeSat-2. The BioCD platform used for these studies could easily be adapted for studies of gravity responses in other cell types, such as bone and muscle cells. The major technical advances already made in this SporeSat grant and the new ones that a Phase 2 project could achieve, will greatly facilitate its adaptation for other gravitational studies in space. The Ceratopteris mutant studies will allow a first-time genetic dissection of the most rapid responses induced in single cells by gravity. A number of earlier studies have documented the importance of Ca2+ currents for polarized growth and for gravitropic growth, however, there are few prior studies that have identified and genetically analyzed the molecular components of these currents. Moreover, while it is generally accepted that Ca2+ is a second messenger that participates in a myriad of plant responses, there is still little known about the molecular identity of plant Ca2+ channels and the proteins that interact with and help regulate these channels (e.g., annexin). The proposed genetic tests of Ca2+ pump and channel function and regulation may not only provide novel insights into how the gravity-directed Ca2+ current originates, but also to further advance an understanding of channel structures. Most of the findings described above were presented in one peer-reviewed, full-length journal publication and/or in three presentations at meetings, which were published as abstracts. These publications are listed in the Bibliography section of this Task Book report.
Bibliography Type:	Description: (Last Updated: 10/16/2023)
Abstracts for Journals and Proceedings	Salmi M, Malcolm H, Cantero-Garcia A, Clark G, Akhavan N, Roux SJ. "Transport activities controlling gravity perception and response in fern spore." Presented at Plant Biology 2016, Austin, TX, July 9-13, 2016. Plant Biology 2016, Austin, TX, July 9-13, 2016. , Jul-2016
Abstracts for Journals and Proceedings	Salmi M, Cantero A, Akhavan, N, Clark G, Roux S. "Genetic approach to testing the role of annexin in mediating the gravity response of Ceratopteris spores." Presented at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Cannon AE, Sabharwal T, Salmi ML, Chittari GK, Annamalai V, Leggett L, Morris H, Slife C, Clark G, Roux SJ. "Two distinct light-induced reactions are needed to promote germination in spores of Ceratopteris richardii." Front Plant Sci. 2023 Jun 2;14:1150199. https://doi.org/10.3389/fpls.2023.1150199 ; PMID: 37332704; PMCID: PMC10272463 , Jun-2023
Articles in Peer-reviewed Journals	Cannon AE, Vanegas DC, Sabharwal T, Salmi ML, Wang J, Clark G, McLamore ES, Roux SJ. "Polarized distribution of extracellular nucleotides promotes gravity-directed polarization of development in spores of Ceratopteris richardii." Front Plant Sci. 2023 Oct 3;14:1265458. https://doi.org/10.3389/fpls.2023.1265458 , Oct-2023
Articles in Peer-reviewed Journals	Park J, Salmi ML, Wan Salim WW, Rademacher A, Wickizer B, Schooley A, Benton J, Cantero A, Argote PF, Ren M, Zhang M, Porterfield DM, Ricco AJ, Roux SJ, Rickus JL. "An autonomous lab on a chip for space flight calibration of gravity-induced transcellular calcium polarization in single-cell fern spores." Lab Chip. 2017 Mar 14;17(6):1095-103. http://dx.doi.org/10.1039/c6lc01370h ; PubMed PMID: 28205656 , Mar-2017